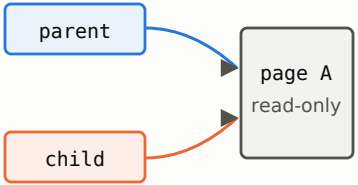
Does fork() copy the parent's memory? Measured answer: no — not until something writes.

1. After fork()

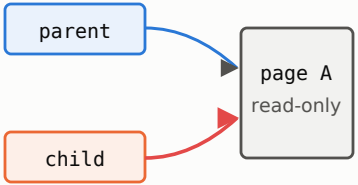
same pages, both marked read-only



one physical page, two owners

2. The child writes

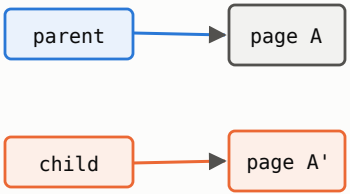
writing a read-only page traps



write → page fault → kernel

3. One page is copied

only the page written; rest shared



now genuinely independent

What we measured — parent holding 256 MB, fork + wait, median of 40 rounds

child exits immediately	**5.50 ms**	nothing copied — only page tables duplicated
child writes every page	**184.48 ms**	every page faulted and copied

forcing the copy costs 33.6× more — and exec would have thrown every copied page away

Figure 4 — Copy-on-write. This is why fork followed immediately by exec is not wasteful: exec discards the whole address space, and almost none of it had been copied.